



Mohammed Reabai J 250701435-2025-CSE ▾

M2

Started on	Saturday, 12 September 2026, 10:00 PM
State	Finished
Completed on	Saturday, 12 September 2026, 10:01 PM
Time taken	54 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      int a[n];
9      int dp[n];
10
11     for (int i = 0; i < n; i++)
12     {
13         scanf("%d", &a[i]);
14         dp[i] = 1;
15     }
16
17     int max = 1;
18
19     for (int i = 1; i < n; i++)
20     {
21         for (int j = 0; j < i; j++)
22         {
23             if (a[i] >= a[j] && dp[i] < dp[j] + 1)
24             {
25                 dp[i] = dp[j] + 1;
26             }
27         }
28
29         if (dp[i] > max)
30             max = dp[i];
31     }
32
33     printf("%d", max);
34
35     return 0;
36 }
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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